

Fractional memory effects in nonlinear waves: Dynamical study of solitons and chaotic attractors

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This paper examines the intricate nonlinear wave behavior characterized by the modified Korteweg–de Vries–Zakharov–Kuznetsov (mKdV–ZK) equation expressed in terms of the M-truncated fractional derivative. To address the involved fractional nonlinear partial differential equation, we use the technique, which efficiently simplifies the nonlinear

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partial differential equation to an ordinary differential equation, enabling the derivation of exact analytical solutions that exhibit bright, dark, and singular wave structures. The characteristics of these solutions are examined via a series of two-dimensional line plots, three-dimensional surface profiles, and comprehensive density maps, giving transparent insights into their physical dynamics and propagation trends. Beyond wave isolated solutions, the nonlinear dynamics are further investigated by recasting the system into an externally forced nonlinear oscillator so that its chaotic regimes can be investigated through time series analyses, phase space portraits, and poincaré diagrams. Results show the presence of multistability, multiple coexisting attractors in certain regimes of parameters, highlighting the profound impact of memory effects on nonlinear wave evolution. This study presents the first chaos and multistability analysis of the M-truncated fractional Korteweg–de Vries–Zakharov–Kuznetsov (KdV–ZK) equation, revealing memory-induced nonlinear dynamics that have not been reported before.

Keywords: Nonlinear partial differential equation; soliton solution; M-truncated fractional derivative; multistability; phase plots.

1. Introduction

Partial differential equations (PDEs), which are derived from intrinsic theories and practical issues, are crucial to the exploration and explanation of the underlying mechanisms of various natural phenomena. PDEs are also a key area in the development of pure mathematics, with the theory of Sobolev spaces being one such renowned [1–5]. Unlike their linear counterparts, nonlinear partial differential equations (NPDEs) include the effects of spatial variation, temporal evolution, delays, and other complicated effects and so provide a more accurate representation of many physical phenomena. In the last several decades, NPDEs have gained a lot of interest because of their potential to simulate various phenomena of interest in scientific fields, including nonlinear optics, plasma physics, signal processing, quantum mechanics, fluid dynamics, and field theory. As a result, the derivation of exact solutions of NPDEs has been an issue of prominent interest, since such solutions reveal significant information on the dynamical properties and inherent mechanisms of the physical systems involved [6–10].

Over the last few years, fractional nonlinear partial differential equations (FNPDEs) have become a dominant extension of traditional NPDEs, involving noninteger-order derivatives to reflect memory effects, hereditary features, and spatial heterogeneity in complicated systems. In contrast to integer-order models, FNPDEs provide an elastic mathematical formulation for modeling processes where the future state depends not just on the current but also on the complete history of the system [11–13]. This makes them highly effective in modeling phenomena in anomalous diffusion, viscoelasticity, biological systems, control theory, and nonlinear wave propagation. The inclusion of fractional derivatives enables a more accurate characterization of long-range interactions and multiscale dynamics, thereby enriching the analytical and numerical study of nonlinear systems and expanding the scope of potential applications in both theoretical and applied sciences [14, 15].

The main aim of the present investigation is to obtain exact solutions to the $(3 + 1)$ -dimensional time-fractional Korteweg–de Vries–Zakharov–Kuznetsov

(KdV–ZK) equation using the M-truncated derivative method. The KdV–ZK model, an extension of the classical KdV one to higher dimensions, is central in the description of nonlinear wave propagation in dispersive media like plasma waves, ion-acoustic waves, and shallow water hydrodynamics. The addition of a fractional time derivative in this model allows for the inclusion of memory effects and nonlocal temporal behaviors that cannot be accounted for by integer-order derivatives. The M-truncated derivative formulation provides a computationally effective and analytically manageable approach for dealing with the fractional-order terms, making the development of closed-form solutions feasible. These closed-form solutions not only grant greater insight into the inherent nonlinear dynamics but also can be used as benchmarks to verify the numerical algorithms used in fractional wave models [16]. Here, we study the $(3 + 1)$ -dimensional time-fractional KdV–ZK equation written in the form

$$D_t^{\alpha,\omega} U + \theta_1 D_x^{2\alpha,\omega} U + \theta_2 D_{xxx}^{3\alpha,\omega} U + \theta_3 (D_{yyx}^{3\alpha,\omega} U + D_{zzx}^{3\alpha,\omega} U) = 0, \quad (1)$$

where $0 < \alpha \leq 1$ is the order in the sense of the M-truncated fractional derivative $D^{\alpha,\omega}(\cdot)$, and $\omega > 0$ is the truncation parameter for the Mittag-Leffler kernel. The parameters θ_1 , θ_2 , and θ_3 define the relative magnitude of dispersive and nonlinear effects along the x , y , and z spatial directions, respectively. Physically, Eq. (1) describes the development of weakly nonlinear ion-acoustic or magnetoacoustic waves in a magnetized plasma with long-wavelength perturbations, in which the fractional derivative in time is a measure of memory and hereditary effects within the medium. The third-order derivatives in x , y , and z describe nonlinear wave steepening and multidirectional dispersion, and the second-order term captures dissipative or diffusive contributions. By including the M-truncated fractional derivative, this model extends the traditional KdV–ZK equation to describe complex physical processes in which traditional integer-order derivatives cannot capture the nonlocal and fractional-order dynamics.

The Korteweg–de Vries (KdV) equation, initially derived for explaining shallow water wave propagation in fluid mechanics, has found itself as a basic model for solitary wave phenomenon and a building block for the investigation of NPDEs [17]. Since then, its relevance has widened far beyond its original fluid-dynamical origin, having relevance in various physical contexts like ion-acoustic and magnetoacoustic waves in plasmas, elastic media longitudinal diffusion waves, and low-temperature thermal excitations in nonlinear crystal lattices. Apart from its wide applicability, the KdV equation has been thoroughly studied from a mathematical structure perspective, such as Painlevé analysis [18], Lax pair representations [19] and Bäcklund transformations [20]. As a multidimensional generalization, the Zakharov–Kuznetsov (ZK) equation broadens the KdV paradigm to multidimensional spaces, providing a more physically realistic description of physical systems where wave propagation takes place in more than one spatial direction. The KdV equation is one of the most fundamental nonlinear evolution equations in mathematical physics and was originally introduced as an integrable model admitting

soliton solutions. Later, the Zakharov–Kuznetsov (ZK) equation was developed as a higher-dimensional generalization of the KdV equation in order to describe multidirectional wave propagation, preserving the dispersive and nonlinear balance of the original KdV model. The modified KdV (mKdV) equation was subsequently introduced to incorporate cubic nonlinearities, leading to richer mathematical structures and solution families compared with the classical KdV equation. The combination of these two extensions produced the mKdV–ZK equation, which serves as a higher-dimensional nonlinear evolution equation with stronger nonlinear interactions.

In recent years, fractional generalizations of KdV-type equations have been actively investigated to extend their mathematical structure by incorporating nonlocal operators and memory effects. Most fractional mKdV and ZK equations reported in the literature are formulated using classical fractional derivatives such as the Caputo or Riemann–Liouville operators. However, these operators introduce strong nonlocality that often complicate both analytical and dynamical investigations. The originality of Eq. (1) lies in constructing the mKdV–ZK equation using the *M-truncated fractional derivative*, which provides a new fractional operator based on the truncated Mittag-Leffler kernel. This formulation preserves key properties of classical differentiation while allowing a controllable fractional order. As a result, Eq. (1) represents a mathematically distinct fractional extension of the mKdV–ZK equation, enabling the derivation of exact traveling-wave solutions and a consistent dynamical system representation that has not been available for previously studied fractional forms of the equation. The KdV–ZK equation in plasma physics describes ion-acoustic wave dynamics for a magnetized plasma system with fluid ions and hot and cold electron populations, and has also been used in fluid dynamics and other nonlinear dispersive systems. The fractional-order KdV–ZK model used in this study includes the M-truncated fractional derivative, which allows for a better representation of wave processes through the incorporation of nonlocal memory effects not accounted for in the conventional integer-order model. The time-fractional modified Korteweg–de Vries–Zakharov–Kuznetsov (mKdV–ZK) equation employed in this study provides a physically meaningful model for nonlinear wave propagation in dispersive and anisotropic media, particularly in plasma physics, ion-acoustic waves, optical fibers, and fluid dynamics. The incorporation of the M-truncated fractional derivative enables the model to capture memory and hereditary effects, which are intrinsic to many real-world materials such as viscoelastic media, plasmas with long-range particle interactions, and optical materials exhibiting delayed nonlinear responses. These memory effects significantly influence wave localization, stability, and the onset of instabilities and chaos. Therefore, the present fractional mKdV–ZK model offers a more realistic framework for describing wave transport, signal propagation, and nonlinear dynamical behavior in complex physical and engineering systems.

Recently, novel solution approaches have been introduced, including Extended direct algebraic method [21], Kudryashov’s method [22, 23], Sardar sub-equation

method [24], Bernoulli sub-ODE method [25, 26], Jacobi's elliptic function [27, 28], mapping method [29–31], the extended hyperbolic function method [32–36], and, ϕ^6 -expansion method [37]. In this paper, we initially use the $(\frac{1}{\varphi(\xi)}, \frac{\varphi'(\xi)}{\varphi(\xi)})$ -expansion method in combination with the M -truncated fractional derivative approach to obtain exact analytical solutions of the time-fractional KdV–ZK equation (1). The M -truncated derivative, rooted in the truncated Mittag-Leffler function, offers a mathematically sound means of including memory effects within the governing equation and thus increasing the physical relevance of the model. By a suitable wave transformation, e.g. the Galilean transformation, the fractional PDE is simplified to an ODE. The simplified ODE is next solved by assuming a finite series expansion in $(\frac{1}{\varphi(\xi)}, \frac{\varphi'(\xi)}{\varphi(\xi)})$, where $\varphi(\xi)$ is a solution of a Riccati-type auxiliary equation. The approach systematically provides different closed-form solutions such as bright and dark solitons, kink waves, and periodic wave patterns.

In the second half of the research, a thorough chaos analysis is performed for the truncated nonlinear dynamical system derived from (11) through the Galilean transformation [14, 38]. The system of first-order ODEs resulting from it is integrated numerically through the fourth-order Runge–Kutta scheme in various parameter regimes, specifically with respect to changes in the external forcing amplitude δ_0 and frequency ϕ . To visualize the rich dynamics, we create three types of plots: (i) three-dimensional phase portraits in order to reveal global attractor patterns, (ii) two-dimensional phase portraits for comparative analysis, and (iii) time series plots to observe oscillatory behavior and onset of chaos. In addition, by keeping system parameters constant and varying initial conditions, we also study the phenomenon of multistability and find the coexistence of multiple stable attractors within the same parameter space. This is an integrated exact-solution and chaos-analysis approach, which not only offers clear analytical wave patterns but also reveals the intrinsic nonlinear dynamics and stability properties of the fractional KdV–ZK system. Although the KdV–ZK equation and its fractional extensions have been widely investigated, existing studies are mainly limited to either integer-order models or classical fractional operators and focus predominantly on deriving analytical solutions or numerical simulations separately. Very few works incorporate memory effects through the M -truncated fractional derivative, and, to the best of our knowledge, no study has simultaneously addressed exact fractional soliton solutions together with chaos, multistability, and Poincaré analysis within the same KdV–ZK framework. This constitutes a clear research gap. This work fills this gap by developing a unified analytical-dynamical approach, allowing us to explore how fractional memory modifies soliton profiles, stability, and chaotic behavior in multidimensional nonlinear wave systems.

The step-by-step workflow of our study is illustrated in Fig. 1, summarizing the process from mathematical formulation to chaos analysis and final conclusions.

The structure of this paper is organized as follows: Section 2 introduces the concept of the M -truncated derivative, establishing the requisite theoretical

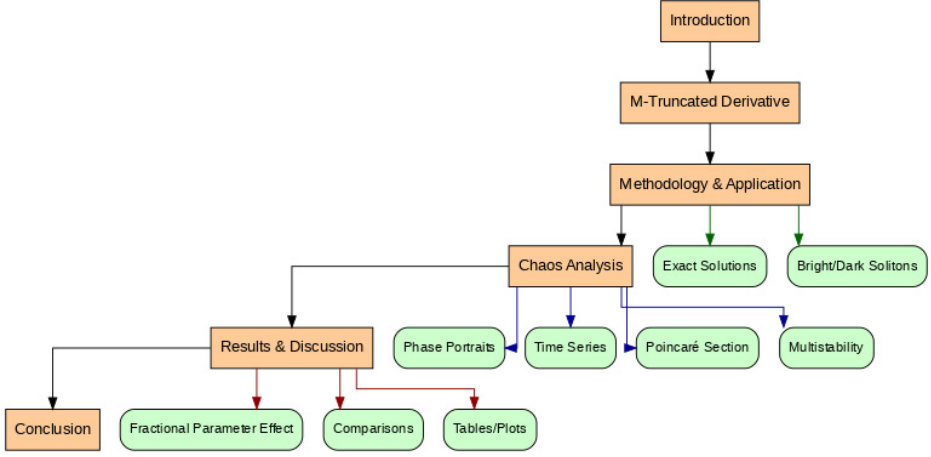


Fig. 1. Flowchart illustrating the methodology and application steps of the proposed work.

background and mathematical framework essential for the analysis to follow. Section 3 provides the detailed methodology, including the particular analytical techniques used and their application to the nonlinear models at hand. Section 4 is reserved for the chaos analysis, wherein the nonlinear dynamical nature of the system is extensively explored via phase portraits, bifurcation diagrams, and other pertinent tools in order to determine chaotic regimes. The findings from these analyses are summarized and interpreted in Sec. 5, along with extensive discussion of the implications and physical significance of the results. Last, Sec. 6 concludes the paper by summarizing the most important contributions, mentioning potential applications, and suggesting directions for future work.

2. M-Truncated Fractional Derivative

The M-truncated fractional derivative (TMD) is a new fractional calculus operator that addresses some drawbacks of current fractional derivatives. The presentation is derived from the use of the truncated Mittag-Leffler function, given by [39]

$$E_{\omega}(z) = \sum_{k=0}^i \frac{z^k}{\Gamma(\omega k + 1)}, \quad \omega > 0. \quad (2)$$

The TMD offers a generalized framework for distinguishing functions incorporating the memory and hereditary effects inherent to fractional calculus.

Definition. Let $f : (0, \infty) \rightarrow \mathbb{R}$. The M-truncated fractional derivative of f of order α , denoted by ${}^iDM_{\alpha, \omega}$, is defined by

$${}^iDM_{\alpha, \omega} f(\tau) = \lim_{\epsilon \rightarrow 0} \frac{f(\tau + iE_{\omega}(\epsilon \tau^{-\alpha})) - f(\tau)}{\epsilon}, \quad (3)$$

where $0 < \alpha \leq 1$, $\omega > 0$, and ${}^iE_\omega(\cdot)$ is the one-parameter truncated Mittag-Leffler function as presented in (1). If f is α -differentiable on some interval $(0, a)$, $a > 0$, and $\lim_{\tau \rightarrow 0+} ({}^iDM_{\alpha,\omega}f(\tau))$ exists, then ${}^iDM_{\alpha,\omega}f(0)$ is defined by this limit. The following properties of the M-truncated fractional derivative are valid:

Theorem. *Let $0 < \alpha \leq 1$, $\omega > 0$, $a, b \in \mathbb{R}$, and f, g be α -differentiable functions at a point $\tau > 0$. Then*

- (1) *Linearity:* ${}^iDM_{\alpha,\omega}(af + bg) = a {}^iDM_{\alpha,\omega}(f) + b {}^iDM_{\alpha,\omega}(g)$.
- (2) *Power rule:* ${}^iDM_{\alpha,\omega}(\tau^\mu) = \mu \tau^{\mu-\alpha}$, $\mu \in \mathbb{R}$.
- (3) *Product rule:* ${}^iDM_{\alpha,\omega}(fg) = f {}^iDM_{\alpha,\omega}(g) + g {}^iDM_{\alpha,\omega}(f)$.
- (4) *Quotient rule:* ${}^iDM_{\alpha,\omega}(\frac{g}{f}) = \frac{g {}^iDM_{\alpha,\omega}(f) - f {}^iDM_{\alpha,\omega}(g)}{f^2}$.
- (5) *If f is differentiable, then* ${}^iDM_{\alpha,\omega}(f)(\tau) = \frac{\tau^{1-\alpha}(\omega)^{1-\alpha}}{\alpha} \frac{df}{d\tau}$.
- (6) *Chain rule:* ${}^iDM_{\alpha,\omega}(f \circ g)(\tau) = f'(g(\tau)) {}^iDM_{\alpha,\omega}(g)(\tau)$.

The M-truncated derivative provides a versatile and efficient mathematical formalism for the solution of fractional PDEs since it maintains compatibility with a variety of traditional calculus properties while also embracing fractional-order dynamics.

3. Description of the $(\frac{1}{\phi(\xi)}, \frac{\phi'(\xi)}{\phi(\xi)})$ Method

The $(\frac{1}{\phi(\xi)}, \frac{\phi'(\xi)}{\phi(\xi)})$ -expansion method is employed in this study because it provides a concise and computationally efficient framework for constructing exact traveling-wave solutions of nonlinear fractional PDEs. Compared with many traditional expansion techniques, this method requires a shorter algebraic procedure and significantly reduces symbolic complexity. More importantly, by selecting different forms of the auxiliary function $\phi(\xi)$ through its governing equation, a wide variety of wave structures can be generated, including bright solitons, dark solitons, kink-type waves, periodic solutions, and singular solutions. This flexibility makes the method particularly suitable for fractional-order models, where memory effects lead to rich and diverse solution behaviors. Consequently, the chosen method allows us to systematically obtain multiple families of physically meaningful solutions within a unified and compact analytical framework.

Consider a general nonlinear evolution equation of the form

$$R(U_{xy}, U_{zt}, D_{i,t}^{\alpha,\beta} U_x, \dots) = 0, \quad (4)$$

where $U(x, y, z, t)$ describes the soliton profile dependent on spatial variable x and time t .

Assume that the solution admits a wave transformation of the form

$$U(x, y, z, t) = U(\xi), \quad \xi = \Gamma \left(\frac{\omega + 1}{\alpha} \right) (ax^\alpha + by^\alpha + cz^\alpha + dt^\alpha), \quad (5)$$

where ξ is a traveling wave coordinate and the term involving the Gamma function $\Gamma(\cdot)$ accounts for the effects of the fractional-order time derivative.

Substituting the transformation from Eq. (5) into Eq. (4), the original fractional partial differential equation is converted into a corresponding ordinary differential equation:

$$T(U, U', U'', U''', \dots) = 0. \quad (6)$$

To solve Eq. (6), we propose a solution of the form

$$U(\xi) = q_0 + \sum_{i=1}^N \left(\frac{q_i + s_i [\varphi'(\xi)]^i}{[\varphi(\xi)]^i} \right), \quad (7)$$

where q_0 , q_i , and s_i ($i = 1, 2, \dots, N$) are constants to be determined. The value of N is identified by applying the method of homogeneous balance between the highest-order derivative and nonlinear terms in Eq. (6).

The auxiliary function $\varphi(\xi)$ in Eq. (7) is assumed to satisfy the differential equation:

$$[\varphi'(\xi)]^2 = \varphi^2(\xi) - \sigma \quad (8)$$

whose general solution is given by

$$\varphi(\xi) = g e^{\xi} + \frac{\sigma}{4g e^{\xi}} \quad (9)$$

with g being an arbitrary integration constant.

By substituting Eqs. (7) and (8) into Eq. (6), a system of algebraic equations is obtained for the constants q_0 , q_i , and s_i . Solving this system provides explicit expressions for the soliton solution.

3.1. Application of $(\frac{1}{\varphi(\xi)}, \frac{\varphi'(\xi)}{\varphi(\xi)})$ Method

Using the wave transformation (5) in (1) and differentiating $U(\xi)$ with respect to x , y , z , and t , and substituting into the governing equation, we obtain

$$dU' + \theta_1 U^2 U' + (a^3 \theta_2 + ab^2 \theta_3 + ac^2 \theta_3) U''' = 0, \quad (10)$$

where primes denote derivatives with respect to ξ . Integrating Eq. (10) once with respect to ξ yields

$$dU + \frac{\theta_1}{3} U^3 + (a^3 \theta_2 + ab^2 \theta_3 + ac^2 \theta_3) U'' = 0. \quad (11)$$

By using the homogeneous balancing rule, we get $N = 1$. From Eq. (7), we assume the following solution:

$$U(\xi) = q_0 + \frac{q_1 + s_1 \phi'(\xi)}{\phi(\xi)}. \quad (12)$$

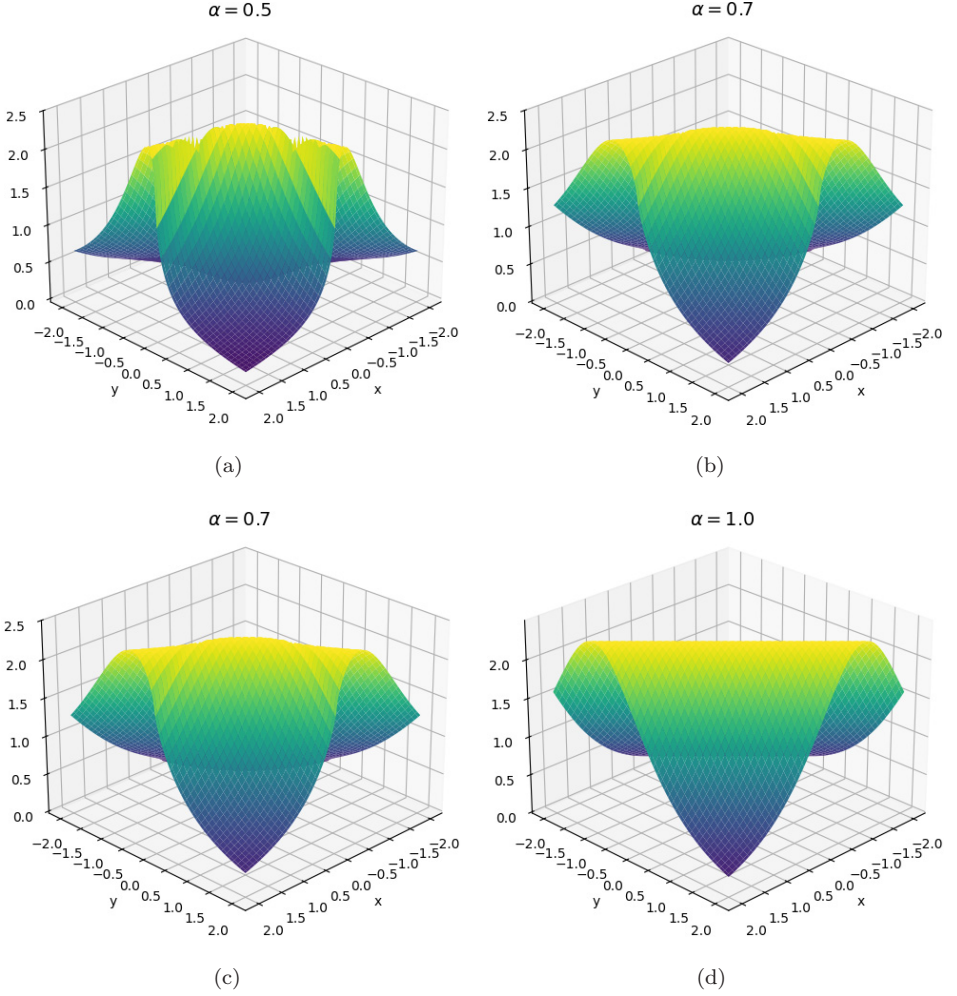


Fig. 2. The dynamical behavior of the bright soliton solution for $U_{1,1}(x, y, z, t)$ at different values of the fractional order α : (a) solution profile for $\alpha = 0.5$; (b) solution profile for $\alpha = 0.7$; (c) solution profile for $\alpha = 0.9$; (d) solution profile for $\alpha = 1.0$. The plots illustrate how the change in α influences the amplitude and shape of the wave structure.

By putting Eqs. (12) and (8) into Eq. (11), we get the algebraic system and by solving this system, the following sets of solutions are obtained:

$$\text{Set : 1} \left\{ q_0 = 0, q_1 = \frac{\sqrt{-6d\sigma}}{\sqrt{a\theta_1}}, s_1 = 0, b = -\frac{\sqrt{a^3(-\theta_2^2) - ac^2\theta_3 - d}}{\sqrt{a\theta_3}} \right\}.$$

By using the above values of constants, we have the following solution:

$$U_1(x, y, z, t) = \frac{4ge^{\xi}\sqrt{-6d\sigma}}{\sqrt{a\theta_1}(4g^2e^{2\xi} + \sigma)}.$$

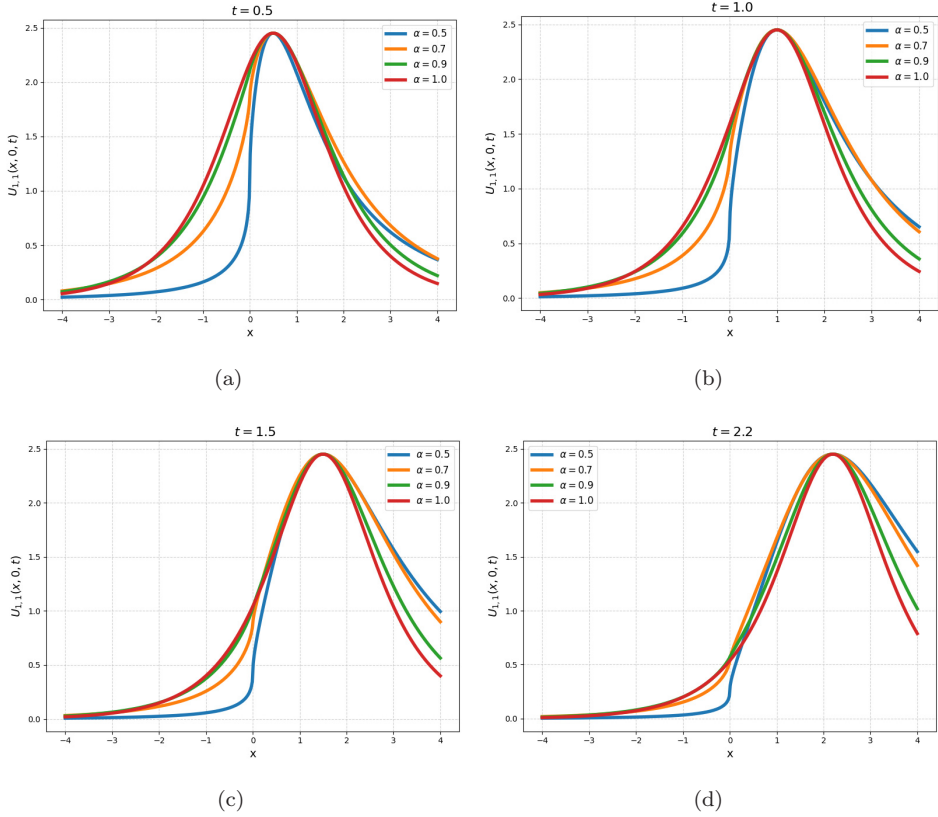


Fig. 3. Comparison of the solution $U_{1,1}(x, 0, t)$ for different values of the fractional order parameter α at fixed time snapshots $t = 0.5$, $t = 1.0$, and $t = 1.5$. Each curve corresponds to a distinct α value, illustrating the influence of α on the amplitude and spatial profile of the solution at the specified time.

By taking $\sigma = \pm 4g^2$, the obtained solutions are given below, respectively.

$$U_{1,1}(x, y, z, t) = \frac{\sqrt{-6d} \operatorname{sech}(\xi)}{\sqrt{a\theta_1}} \quad d < 0, \quad a\theta_1 > 0.$$

$$U_{1,2}(x, y, z, t) = \frac{\sqrt{-6d} \operatorname{csch}(\xi)}{\sqrt{a\theta_1}} \quad d < 0, \quad a\theta_1 > 0.$$

$$\text{Set : } 2 \left\{ q_0 = 0, q_1 = 0, s_1 = -\frac{\sqrt{3d}}{\sqrt{a\theta_1}}, b = \frac{\sqrt{-2a^3\theta_2^2 - 2ac^2\theta_3 + d}}{\sqrt{2a\theta_3}} \right\}.$$

By using the above values of constants, we have the following solution:

$$U_2(x, y, z, t) = \frac{\sqrt{-3d}(\sigma - 4g^2 e^{2\xi})}{\sqrt{a\theta_1}(4g^2 e^{2\xi} + \sigma)}.$$

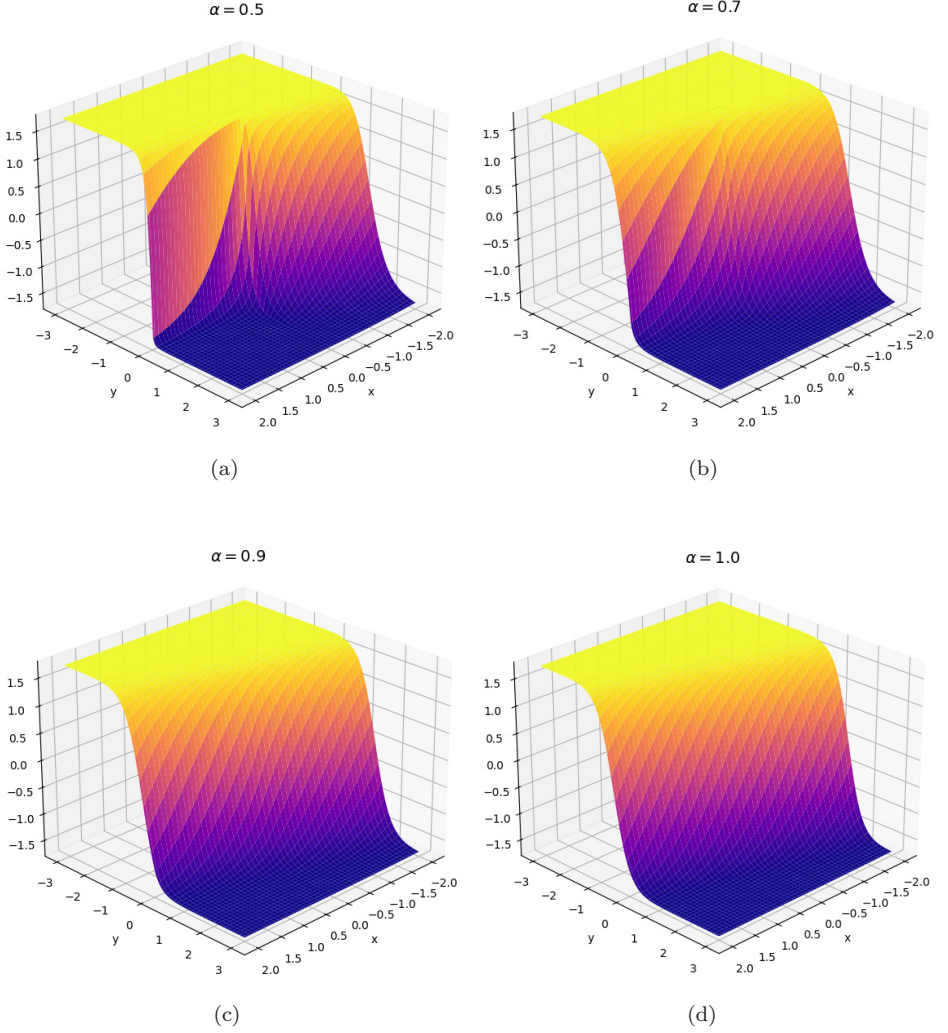


Fig. 4. The dynamical behavior of the kink soliton solution for $U_{2,1}(x, y, z, t)$ at different values of the fractional order α : (a) solution profile for $\alpha = 0.5$; (b) solution profile for $\alpha = 0.7$; (c) solution profile for $\alpha = 0.9$; (d) solution profile for $\alpha = 1.0$. The plots illustrate how the change in α influences the amplitude and shape of the wave structure.

By taking $\sigma = \pm 4g^2$, the obtained solutions are given below, respectively.

$$U_{2,1}(x, y, z, t) = -\frac{\sqrt{3d} \tanh(\xi)}{\sqrt{a\theta_1}}, \quad d > 0, \quad a\theta_1 > 0.$$

$$U_{2,2}(x, y, z, t) = -\frac{\sqrt{3d} \coth(\xi)}{\sqrt{a\theta_1}}, \quad d > 0, \quad a\theta_1 > 0.$$

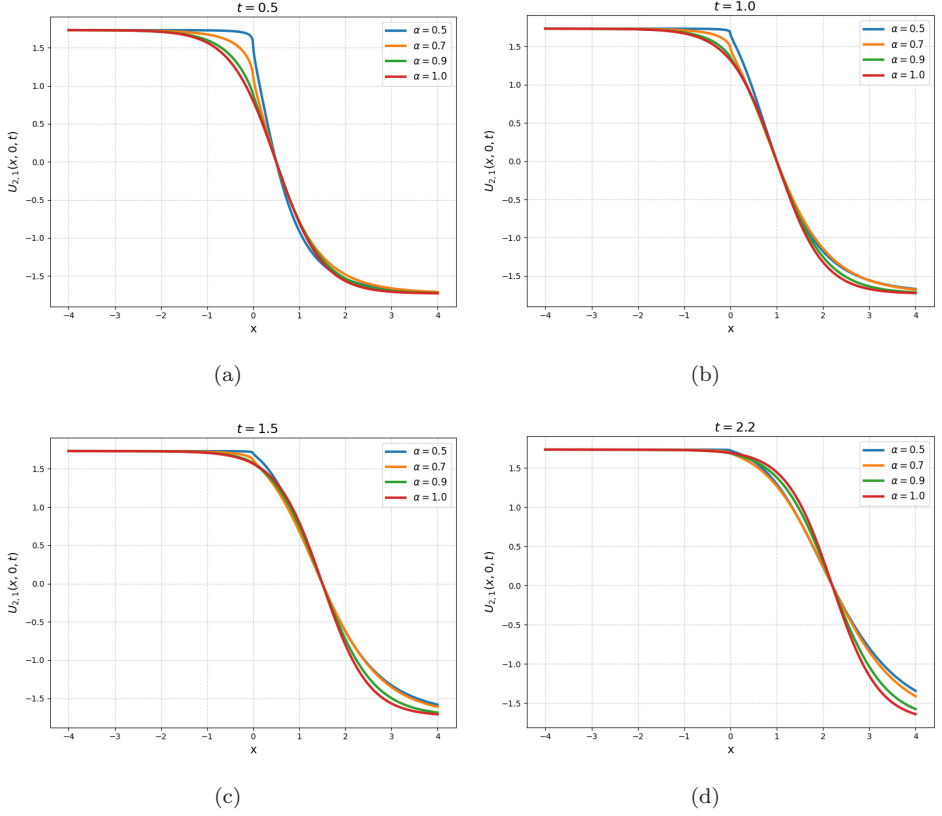


Fig. 5. Comparison of the solution $U_{2,1}(x, 0, t)$ for different values of the fractional-order parameter α at fixed time snapshots $t = 0.5$, $t = 1.0$, and $t = 1.5$. Each curve corresponds to a distinct α value, illustrating the influence of α on the amplitude and spatial profile of the solution at the specified time.

4. Chaos Analysis

Applying the Galilean transformation, Eq. (11) can be rewritten as the following dynamical system:

$$\begin{aligned} \frac{dU}{d\xi} &= Q, \\ \frac{dQ}{d\xi} &= r_0 U + r_1 U^3 + \delta_0 \sin(\phi\xi), \end{aligned} \quad (13)$$

where

$$r_0 = \frac{-d}{a^3\theta_2 + ab^2\theta_3 + ac^2\theta_3}, \quad r_1 = \frac{-\theta_1}{3(a^3\theta_2 + ab^2\theta_3 + ac^2\theta_3)}.$$

The system (13) is solved numerically by the fourth-order Runge–Kutta method with constant step size. Phase portraits and relevant time series are plotted for

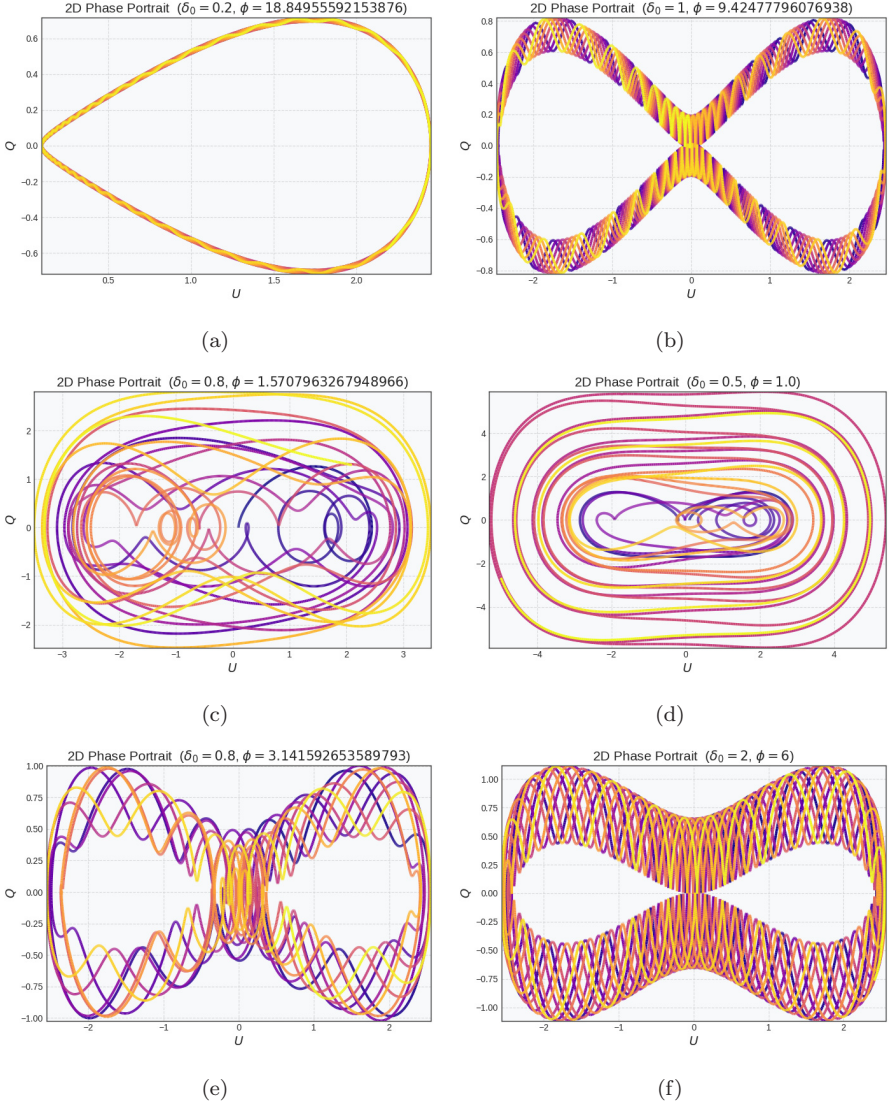


Fig. 6. Phase portraits of the system (13) for various values of δ_0 and ϕ , showing the variation of dynamical behavior as the external forcing parameters change.

different pairings of δ_0 and ϕ , while other parameters are held constant as $a = b = c = \theta_1 = \theta_2 = \theta_3 = 1$, $d = -1$, $U(0) = 0.1$, $Q(0) = 0$. We explore the impact of changing δ_0 and ϕ while keeping other parameters fixed.

Figures 6 and 7 present a collection of two-dimensional phase portraits (U, Q) for various (δ_0, ϕ) combinations. Each subplot includes the system trajectory in phase space after removing the initial transients. The figures evidently show how increasing the forcing amplitude or detuning the forcing frequency can lead the system away from regular periodic motion and towards more random, complex behavior.

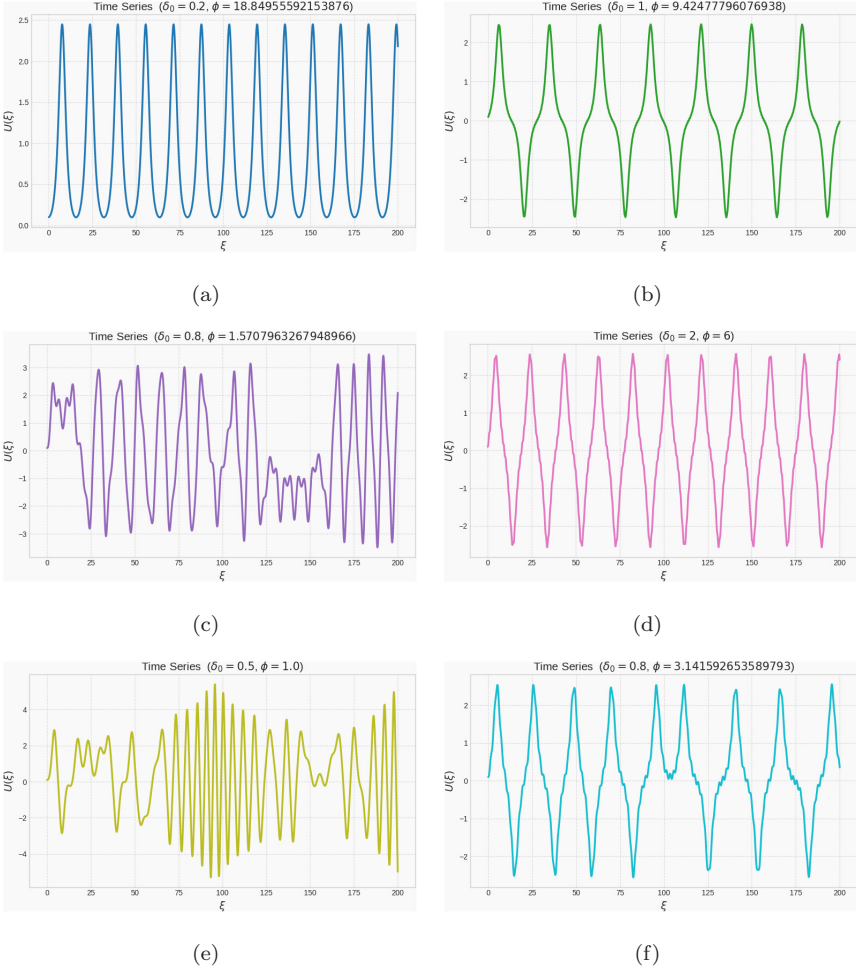


Fig. 7. Time series plots of $U(\xi)$ corresponding to the phase portraits in Fig. 6, illustrating the time evolution of the system for different combinations of δ_0 and ϕ .

At small forcing amplitudes, the trajectories trace out smooth closed curves, which corresponds to periodic or quasi-periodic behavior. With increasing δ_0 , the phase portraits get progressively more densely filled, signifying the emergence of chaotic oscillations.

4.1. Multistability analysis

In nonlinear dynamical systems, *multistability* is the coexistence of various long-term behaviors for the same parameters of the system but different initial conditions.

To study this phenomenon in our model, we take Eq. (13) with fixed parameters

$$a = b = c = \theta_1 = \theta_2 = \theta_3 = 1.0, \quad d = -1.0, \quad \delta_0 = 0.8, \quad \phi = \pi, 3\pi.$$

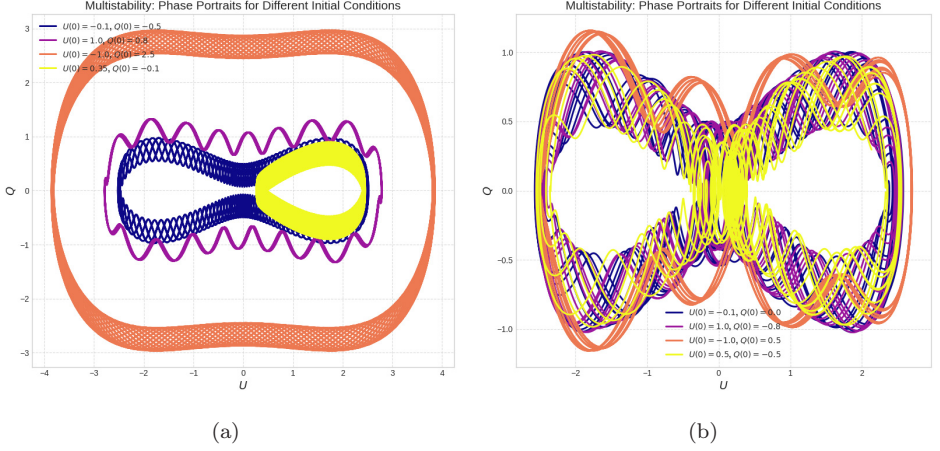


Fig. 8. Multistability in the system: phase portraits for different sets of initial conditions with fixed parameters. Distinct trajectories in the same parameter regime confirm the coexistence of multiple stable behaviors.

To analyze sensitivity to initial conditions, simulations are run for four different sets: The obtained 2D phase portraits are presented in Fig. 8. Each curve is the trajectory of (U, Q) in phase space for a fixed initial condition, and the trajectories are colored for simplicity. The different attractors in the same parameter range prove the coexistence of multiple stable solutions and thus confirm the occurrence of multistability in the system.

4.2. Analysis of Poincaré section

To better study the intricate dynamics of system (13), we use the Poincaré section method, which creates a discrete mapping of the continuous trajectory by sampling the phase space at equal intervals equal to the period of driving $T = \frac{2\pi}{\phi}$. This simplifies the dimensionality of the system, allowing easier visualization of the long-term behavior and determination of periodic, quasi-periodic, or chaotic motion. Numerical integration of system (13) is done by the fourth-order Runge–Kutta method with fixed step size $h = 0.005$, over a long enough time interval to exclude transient effects. The parameters are kept constant as $a = b = c = \theta_1 = \theta_2 = \theta_3 = 1$, $d = -1$, $U(0) = 0.1$, $Q(0) = 0.0$. The forcing parameters are being changed externally for two typical cases:

For every case, two plots are produced: (i) the complete continuous trajectory in the (U, Q) phase plane, and (ii) the same trajectory with the discrete Poincaré points superimposed. The latter are computed by sampling (U, Q) once per driving period after the first 120 forcing cycles are discarded in order to eliminate transient effects. In the first example ($\delta_0 = 0.8, \phi = 5\pi$), the Poincaré section shows a collection of points along closed invariant curves, marking quasi-periodic motion. The smooth trajectory densely covers a torus-shaped area in phase space, guaranteeing the

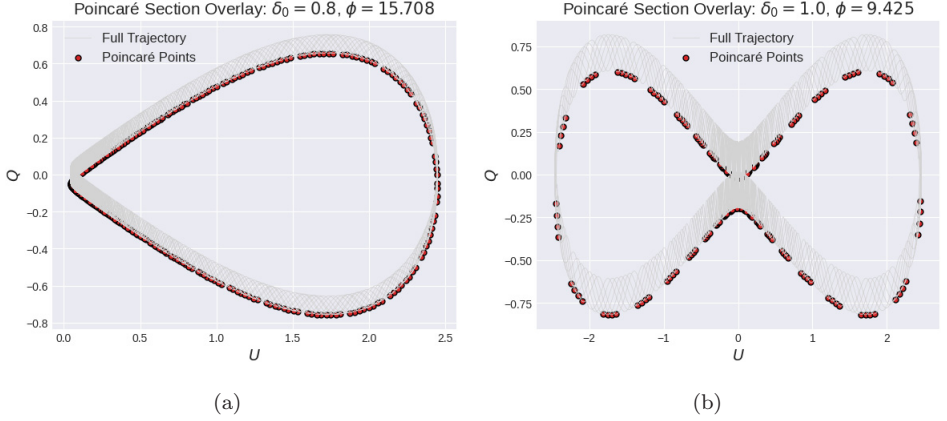


Fig. 9. Comparison between continuous phase portraits and corresponding Poincaré sections of system (13) for (a) $\delta_0 = 0.8, \phi = 5\pi$ showing periodic motion and (b) $\delta_0 = 1.0, \phi = 3\pi$ showing chaotic dynamics.

two incommensurate frequencies in the system. Conversely, the second example ($\delta_0 = 1.0, \phi = 3\pi$) has a scattered and irregular point distribution of Poincaré points with no apparent smooth structure. It is typical of chaotic dynamics where nearby trajectories bifurcate exponentially and the system samples a fractal subset of the phase space. The relation between the continuous phase portraits and their respective Poincaré sections not only confirms the numerical integration but also gives a compact diagnostic device for separating regular and chaotic regimes under various external forcing conditions.

5. Results and Discussion

In this work, we investigate the influence of fractional-order parameter α and time evolution on the soliton profiles calculated via the presented approach, and then present in-depth analysis of the nonlinear dynamical properties of the reduced system (13) under periodic external forcing. We consider bright and dark soliton properties, fractional-order sensitivity, and chaotic responses, including Poincaré maps for characteristic parameter sets.

Here, we discuss the effect of the fractional-order parameter α and time t on the soliton profiles that have been obtained. All simulations have been done with the following parameters: $a = 1.0, b = 1.0, c = 1.0, \theta_1 = 1.0, d = -1.0, \omega = 1.0$. The space coordinate y is held constant at $y = 0$ to simplify the solution to a two-dimensional profile along the x -axis. The fractional exponents are $\alpha = 0.5, 0.7, 0.9, 1.0$, while the time parameter t changes for varying snapshots. Figures 2 and 3 relate to the bright soliton profiles. These show a localized peak with $x = 0$ at its center. For smaller α values (e.g. $\alpha = 0.5$), the solitons are broader, meaning slower spatial decay, while greater α values (e.g. $\alpha = 1.0$) yield narrower and more compact peaks. Raising t slightly displaces and widens the profiles because of the

nonlinear dispersion effects included in the fractional model. Figures 4 and 5 are the dark soliton profiles. They consist of a localized dip (minimum in intensity) on a constant background, a phase shift in the wave field instead of a localized pulse. The profile is wider with a more gradual transition between plateaus when $\alpha = 0.5$, while for $\alpha = 1.0$, the central dip is more localized and sharper.

The numerical study demonstrates that the system governed by Eq. (13) has rich dynamical behavior as functions of forcing parameters (δ_0, ϕ) and the initial conditions selection. For chaos analysis, phase portraits (Fig. 6) and corresponding time series plots (Fig. 7) verify the onset of chaotic behavior, manifested as nonperiodic, irregular trajectories both in phase space and temporal development. The change in (δ_0, ϕ) results in discernible differences in the attractor structure, reflecting sensitive dependence on system parameters. In multistability, several discrete attractors were observed for a single set of parameters but varied initial conditions as given in Fig. 8. This illustrates the coexistence of various stable solutions where the long-term response of the system strongly depends on the initial state. This type of behavior underscores the significance of initial condition sensitivity to forecast the system's evolution.

In order to further describe the system's dynamics, Poincaré maps are formed by phase space (U, Q) sampled stroboscopically at the external forcing period $T = 2\pi/\phi$. Representative parameter sets are chosen to be $(\delta_0, \phi) = (0.8, 5\pi)$ and $(1.0, 3\pi)$. In Fig. 9, the left panels indicate the complete phase trajectories and the right panels plot the corresponding Poincaré points over them. Periodic motion appears as a few discrete points, quasi-periodic motion results in closed invariant curves, and chaotic motion gives scattered points covering areas of the phase plane.

5.1. Memory effects in the fractional mKdV–ZK equation

One of the most interesting aspects of fractional calculus, and specifically the M-truncated fractional derivative used here, is the ability to include memory and hereditary effects into the dynamics of nonlinear wave propagation. This is in contrast to conventional integer-order derivatives, which depict systems in terms of their state at an instant only, while fractional derivatives take into account all of the past history of the system, essentially imparting temporal nonlocality into the model.

For the case of the mKdV–ZK equation, such memory has a key impact on soliton dynamics and chaotic behaviors. The order parameter determines how much memory exists: Smaller values for order focus on the long-term contributions from the past, whereas values near the integer boundary decrease the impact of the past and restore more traditional, memoryless behavior.

Physically, memory is evident in the persistence and stability of nonlinear wave structures. For instance: Soliton width and amplitude become adjustable with order, enabling the same medium to accommodate wider, more smooth solitons or steeper, high-amplitude solitons based on its memory depth. Propagation stability is affected

by hereditary effects, such that earlier states can dampen instabilities and preserve soliton integrity over long distances of transmission — a particularly valuable characteristic for engineering and telecommunication applications. Turbulent regimes exhibit increased richness in fractional memory, where there are several coexisting attractors and multistability since the system reacts to current forcing but also carries imprints of its past dynamics. Practically, the inclusion of memory in nonlinear models like the mKdV–ZK equation makes them more accurate descriptions of physical and engineering systems. Most real processes — e.g. wave transport in plasmas, viscoelastic media, optical fibers, and even climate dynamics — are intrinsically nonlocal in time. By adjusting the fractional order, engineers and scientists can simulate how history-dependent dynamics change wave speed, shape, and stability, opening up new prospects for control and optimization.

Therefore, memory effects enshrined through the M-truncated fractional derivative not only enrich the mathematical scenario of soliton and chaotic dynamics but also open up a flexible and potent tool to design robust, reconfigurable, and fault-tolerant engineering systems.

5.2. Future directions

Although this study provides a comprehensive analytical and dynamical investigation of the M-truncated fractional mKdV–ZK equation, several important directions remain open for future research. One natural extension is to develop and analyze the *stochastic* version of the fractional mKdV–ZK equation in order to examine the influence of random perturbations and noise on soliton stability and chaotic dynamics. In addition, high-accuracy *numerical simulations* such as spectral, finite difference, or finite element methods may be employed to further validate the analytical solutions and to explore regimes where closed-form solutions are not available. From a dynamical systems perspective, future studies may also investigate other quantitative indicators of chaos, such as Lyapunov exponents, bifurcation diagrams, and entropy measures, to provide a deeper understanding of the transition routes to chaos and the robustness of multistable states. These extensions would further enrich the theoretical framework and enhance the predictive power of fractional nonlinear wave models.

6. Conclusion

In this paper, we have studied the nonlinear dynamical characteristics of the considered fractional-order system using the $(\frac{1}{\varphi(\xi)}, \frac{\varphi'(\xi)}{\varphi(\xi)})$ method. This analytical technique allowed obtaining exact bright and dark soliton solutions, whose profiles were examined systematically using 3D surface plots and 2D comparative graphs for different fractional orders α and time points t . The outcomes definitely exhibit the impact of fractional-order effects on soliton amplitude, width, and localization. Aside from the soliton analysis, a rigorous chaos study was conducted by converting the governing equation through the Galilean transformation. Numerical simulations

for various forcing parameters (δ_0, ϕ) yielded varied 2D phase portraits and time series plots, showing transitions from periodic to chaotic motion, thus substantiating the system's sensitivity to parameter changes. In addition, multistability was investigated by changing the initial conditions under constant parameters. The occurrence of more than one coexisting attractor for the same parameter values shows the coexistence of different stable states, high sensitivity to initial conditions, and the possibility of complex dynamical responses. The analytical results demonstrate that the M-truncated fractional mKdV–ZK equation admits a rich family of exact traveling-wave solutions, including bright solitons, dark solitons, and singular structures. It is shown that the fractional order α plays a crucial role in controlling the width, amplitude, and localization of these waves, with smaller values of α producing broader and smoother profiles, while values approaching unity generate more localized and sharper solitons. This confirms that fractional memory significantly alters the wave morphology compared with the classical integer-order case. The dynamical analysis further reveals that the reduced fractional system exhibits complex nonlinear behavior under external forcing. Transitions from periodic to chaotic motion are observed as the forcing amplitude and frequency vary, and multiple coexisting attractors are detected for the same parameter values, confirming the presence of multistability. The Poincaré sections clearly distinguish regular and chaotic regimes, providing strong evidence of sensitive dependence on initial conditions. These results collectively show that the inclusion of M-truncated fractional memory not only modifies soliton structures but also profoundly enriches the underlying nonlinear dynamics of the mKdV–ZK equation. The originality of this study lies in combining the M-truncated fractional derivative with an exact solution technique and nonlinear dynamical analysis in a single framework. This allowed us, for the first time, to investigate how fractional memory simultaneously influences soliton structures, chaotic dynamics, and multistability in the fractional KdV–ZK model. The results demonstrate that memory effects provide an additional control mechanism over wave localization and stability, which is not available in classical integer-order or standard fractional formulations.

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